

Naga Raju Vadelli

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Professional Summary

- Data Engineer with 5+ years of experience in designing, developing, and deploying scalable data solutions across Azure, AWS, and Hadoop ecosystems.
- Proven expertise in data integration, ETL/ELT pipeline design, and end-to-end data lifecycle management, including ingestion, transformation, validation, and analytics enablement.
- Hands-on experience in migrating on-premise SQL databases to Azure Data Lake, Azure Synapse, and Azure Databricks using Azure Data Factory and Spark.
- Experienced in Azure Data Engineering, building end-to-end pipelines using ADF, ADLS Gen2, Synapse, and Azure SQL, with robust security, automation, and monitoring.
- Worked with AWS services including S3, EC2, Redshift, Glue, Athena, and EMR, integrating them into cloud-native pipelines for enterprise-scale analytics.
- Extensive experience in Databricks, developing Delta Lake architectures, building PySpark and Scala notebooks for large-scale ETL, implementing Unity Catalog for governance, and optimizing cluster performance and job orchestration for cost efficiency.
- Skilled in Snowflake data warehousing, leveraging virtual warehouses, secure data sharing, clustering, and performance tuning for enterprise workloads.
- Strong command of Big Data technologies including Hadoop, Spark, Kafka, HBase, Hive, and Delta Lake, with experience optimizing data pipelines for distributed processing.
- Proficient in Python (Pandas, NumPy, PySpark, Scikit-learn) for data wrangling, statistical analysis, and ML model preparation.
- Experienced in data quality and observability frameworks such as Great Expectations and Deequ, ensuring accuracy, compliance, and trust in enterprise datasets.
- Experienced in containerization and CI/CD automation using Jenkins, Docker, Kubernetes, and Azure DevOps to streamline data workflows and deployment processes.
- Adept at data modeling (dimensional and snowflake schemas), data governance, and performance tuning across relational and NoSQL databases (SQL Server, MongoDB, Cassandra, DynamoDB).
- Strong background in cloud-native analytics platforms (Databricks, Azure Synapse, AWS EMR, Airflow, Athena) with a focus on cost-efficient, high-performance data solutions.
- Collaborative team player with a track record of modernizing data ecosystems, improving data reliability, and enabling advanced analytics and business intelligence initiatives.

Technical Skills

Programming Languages: Python (Pandas, NumPy, PySpark), Scala, SQL, Java, Bash/Shell scripting

Data Engineering: Apache Spark, Apache Kafka, Apache NiFi

ETL & Orchestration: Azure Data Factory, Talend, Informatica, Alteryx, Apache Airflow, Luigi, SSIS

Big Data Technologies: Hadoop (HDFS, MapReduce, Hive, Pig), Databricks, Snowflake, Redshift, Impala

Cloud Platforms: AWS (S3, Redshift, Glue, EMR, Lambda, Athena, RDS), Azure (Data Lake, Synapse, Databricks, Event Hub), GCP (BigQuery)

Databases: PostgreSQL, MySQL, SQL Server, Oracle, MongoDB, Cassandra, DynamoDB, Redis

Data Modeling & Warehousing: Star/Snowflake schema design, Data Vault 2.0, Dimensional modeling

DevOps & Automation: Git/GitHub/GitLab, Jenkins, Docker, Kubernetes, Terraform, CloudFormation, Azure DevOps

Data Quality & Governance: Great Expectations, Deequ, Soda SQL, Apache Ranger, AWS Lake Formation

Professional Experience

Data Engineer, Centene Corporation

July 2024 – Present

- Designed and executed scalable data transformation pipelines using Apache Spark, PySpark, Scala, and Python for processing large datasets efficiently.
- Managed and optimized Databricks clusters, implementing autoscaling, job scheduling, and performance tuning to minimize costs and enhance processing efficiency.
- Designed and implemented Snowflake data models to enable efficient data storage, retrieval, and reporting for business stakeholders.
- Maintained and optimized Databases and AWS services, including Amazon S3, Redshift, RDS, and DynamoDB, ensuring high availability and data consistency.
- Configured Snowflake performance optimizations, including query tuning, warehouse sizing, and clustering keys to improve efficiency and cost management.
- Improved data extraction, transformation, and processing using Spark-SQL, AWS Athena, and Glue, optimizing query performance for real-time and batch processing.
- Designed and enforced data security policies, encryption mechanisms, and role-based access controls (RBAC) on Databricks to ensure compliance with GDPR, CCPA, and HIPAA regulations.
- Developed real-time data streaming applications using Apache Kafka, enabling event-driven architectures for real-time analytics and processing.
- Integrated AWS Lambda and Step Functions to orchestrate and automate serverless data workflows, reducing manual intervention and improving operational efficiency.
- Leveraged AWS Glue and Lake Formation to catalog, partition, and manage large-scale datasets stored in AWS S3, improving discoverability and query performance.
- Performed advanced data mining, anomaly detection, and predictive analytics using ML models in PySpark and Databricks MLflow, contributing to a 20% reduction in operational costs.
- Designed and optimized data lake architecture on AWS, ensuring scalable storage and efficient data retrieval for analytical and reporting purposes.
- Implemented automated data validation using Great Expectations to ensure pipeline accuracy and data reliability.
- Collaborated with data scientists, analysts, and stakeholders to translate business needs into scalable data solutions, enhancing data accessibility and decision-making.

Big Data Engineer, Warner Bros

Nov 2021 – July 2023

- Designed and developed large-scale distributed data processing solutions using Hadoop ecosystem, leveraging HDFS, MapReduce, and YARN for efficient storage and processing.
- Developed and optimized Hive-based ETL pipelines, improving query performance through partitioning, bucketing, and indexing techniques.
- Implemented real-time and batch data processing workflows using Apache Spark and Hive, ensuring efficient data transformation and analytics.
- Managed data governance and security controls within Snowflake, ensuring compliance through RBAC, masking policies, and secure access.
- Automated data provisioning and monitoring tasks in Snowflake using scripts and orchestration tools to streamline daily operations.
- Worked extensively with Hadoop MapReduce for processing high-volume structured and unstructured data, optimizing job execution and resource utilization.
- Configured and managed Kafka topics and KSQL queries, enabling real-time streaming and event-driven architectures for business-critical applications.
- Performed performance tuning on Hadoop clusters, optimizing job scheduling with YARN Resource Manager and dynamic partitioning in Hive.
- Automated deployment and monitoring of Hadoop-based data pipelines using Jenkins and AWS CloudWatch, improving system reliability and reducing downtime.

- Created interactive Tableau dashboards, integrating Hadoop and AWS Redshift data sources for real-time business intelligence insights.
- Managed Teradata workloads, optimizing queries and database structures to improve storage efficiency and processing speed.

Azure Data Engineer, Ivy Comp Tech

May 2019 – Oct 2021

- Developed and refined data pipelines in Azure Data Factory and Spark, improving analytics for marketing and mortgage operations. Employed Spark SQL for effective data streaming and batch processing.
- Directed the migration of SSIS packages and database procedures to Azure, enhancing data governance and compliance. Streamlined data lineage and integration processes.
- Engineered RESTful APIs and integrated Java applications, managing projects with Jira. Set up automated monitoring via OpenShift and Jenkins to optimize CI/CD workflows.
- Led design and execution of ETL pipelines via Azure Data Factory, reducing data processing times by 40% by integrating multiple data sources into Azure Data Lake.
- Implemented Kafka for JSON data streaming and developed Spark-Streaming applications to push data to Redshift. Utilized Snowflake and Power BI for sophisticated data analysis.
- Developed custom Databricks notebooks and libraries in Python and Scala to automate data cleansing, transformation, and visualization tasks, enhancing operational efficiency.
- Conducted troubleshooting and debugging of Teradata utilities to resolve issues and enhance system performance.
- Involved in continuous Integration of applications using Jenkins.
- Utilized Azure Databricks for scalable processing of large datasets, optimizing Spark configurations to enhance performance and efficiency.
- Reviewed and optimized Azure resource utilization, achieving a 20% reduction in cloud expenditures.
- Configured security settings and compliance checks in Azure Data Lake, enforcing robust data governance and privacy standards.

Certifications

- Databricks Certified Databricks Fundamentals Accreditation
- Databricks Certified Data Engineer Associate
- Oracle Certified Generative AI Professional
- Oracle Certified Data Science Professional
- IBM Data Science Professional Certificate by Coursera

Education

Texas Tech University, Master of Science in Computer Science

Aug 2023 – May 2025